



Z(5th Sm.)-Computer Sc.-G/SEC-A-1 & SEC-A-2/CBCS

2023

COMPUTER SCIENCE — GENERAL

Paper : SEC-A-1 and SEC-A-2

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Paper : SEC-A-1

(Communication, Computer Network and Internet)

Full Marks : 80

Answer *question nos. 1 and 2* and *any four* questions from the rest.

1. Answer *any ten* questions :

2×10

- (a) What is network topology?
- (b) What is channel capacity?
- (c) Distinguish between periodic and non-periodic signal.
- (d) Differentiate between bit rate and baud rate.
- (e) Define Bandwidth of a medium.
- (f) State Nyquist theorem.
- (g) What do you mean by network protocol?
- (h) What is composite signal?
- (i) What is VSAT?
- (j) State two advantages of Co-axial cable over UTP.
- (k) Define SNR of a signal.
- (l) What is the function of modem?
- (m) Why do we need DNS?
- (n) What is IP address?
- (o) What do you mean by Internet Browser? Give example.

2. Write short notes on *any four* of the following :

5×4

- (a) Microwave Communication
- (b) AM and FM radio
- (c) ISDN

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- (d) FSK
 - (e) TCP/IP Network model
 - (f) Voice and video conferencing.
3. (a) Describe the concept of simplex, half-duplex and full duplex communication with suitable examples.
(b) Describe mesh topology with proper diagram and also state its advantages and disadvantages. 6+4
4. (a) State any two functionalities of physical layer and data link layer of the OSI model.
(b) State the advantages of optical fiber cable.
(c) Compare between LAN and MAN. (2+2)+3+3
5. (a) Describe FSK and PSK with examples.
(b) Explain the concept of QAM. (3+3)+4
6. (a) What is multiplexing?
(b) Describe TDM and FDM.
(c) Explain the function of router. 2+(3+3)+2
7. (a) What is line coding? What is unipolar encoding?
(b) Describe Manchester encoding and differential Manchester encoding with suitable example.
(c) What is Pulse Code Modulation (PCM)? (1+1)+(3+3)+2
8. (a) Describe ADSL.
(b) What is URL?
(c) Explain classful IP addresses. 5+2+3
9. (a) What is WWW?
(b) What do you mean by ISP?
(c) Explain Post Office Protocol, version3 (POP3). Briefly explain how Internet Mail Access Protocol, version4 (IMAP4) is more powerful than POP3. 2+2+(4+2)